

V6.2 · Continuous Biochemical Theme Space

Coordinates, not classes

Objective	Replace theme classification with a continuous, non-negative theme vector, and show that chemistry organises continuously.
Method	$\text{theme}(x) = S^T M^T \text{coord}(x) \in \mathbb{R}^6_+$, stored as theme_embedding.npy. Visualised by PCA and UMAP; distances measured by cosine.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Parts 4 & 8 — chemistry organises continuously, not into discrete bins

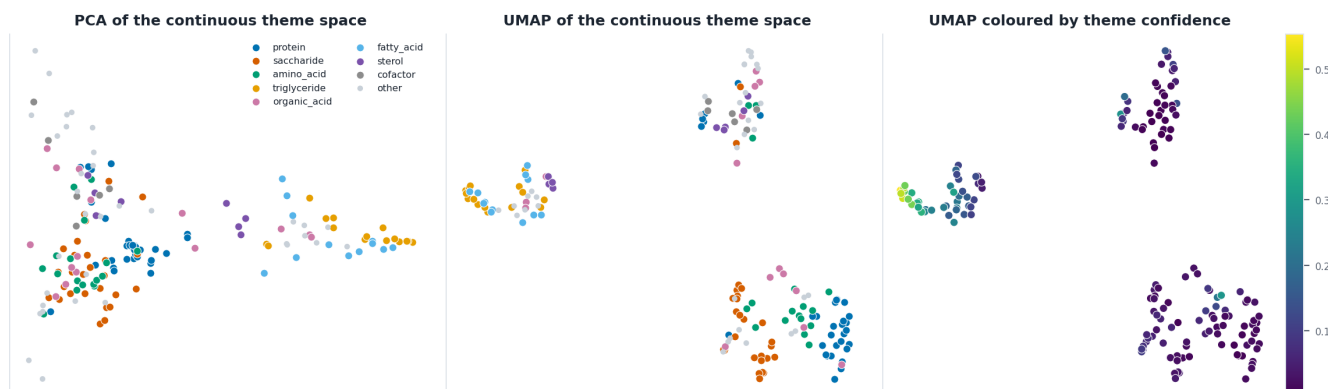


Figure 1 — PCA and UMAP of the continuous theme space, coloured by chemical family and by theme confidence.

Results

embedding dimension	6
themes	Nucleic chemistry, Carbohydrate chemistry, Protein chemistry, Lipid chemistry, Energy metabolism, Redox / pigments
PCA explained variance	0.570, 0.184, 0.137, 0.069
first two PCs	75.5% of the variance
mean pairwise cosine distance	0.324
UMAP available	True

Interpretation

Two principal components carry 75% of the theme-space variance, so the six-dimensional chemical state is close to a surface, not a cloud of discrete bins. In the UMAP the lipid analytes (triglycerides and fatty acids) form a single dense arm — consistent with their near-perfect recovery — while the amino acids and organic acids occupy a diffuse central region, which is exactly where the theme layer is least reliable.

A continuous coordinate is what makes cross-substrate transfer expressible at all: a class label cannot be partially preserved, but a coordinate can move a measurable distance.

Limitations

- UMAP is a visualisation, not a metric; all quantitative statements use the cosine distance in the raw theme space.
- The embedding is unnormalised for spectral intensity, so an analyte with more total motif mass sits further from the origin.
- 167 analytes is small for a manifold claim; the structure is suggestive, not established.

Recommendations

- Use the raw theme vector, not the UMAP coordinates, for any downstream computation.
- Re-examine the manifold once biological cohorts are projected — the interesting question is whether disease states move along it.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.